

Online Supplement - SQPAD

Contents

1 Approach	1
2 SQPAD examples	2
2.1 Simple simulated data	2
2.2 Using the convolution model	5
2.3 Field data example	5
3 Synthetic simulations	7
4 Estimating parameters from simulations	8
5 Field data analysis	9
6 Simulation results	10
6.1 Mean distance vs. average detection probability	16
7 Field data results	19
8 Session info	22

This document is the online supplement for the manuscript *Single bin QPAD (SQPAD) approach for robust analysis of point count data with detection error*.

1 Approach

The following files can be found in this bundle:

- README.md: instructions for running R scripts on a remote server
- 00-functions-sqpad.R: functions implementing the SQPAD estimating procedures (these are also part of the detect package)
- 01-functions-helpers.R: functions for reproducing the analyses
- 02-bsims-simulations.R: synthetic simulation setups
- 03-bsims-estimation.R: estimating parameters from synthetic simulations
- 04-field-data.R: field data analysis using SQPAD
- 05-summaries.R: summarizing synthetic simulation and field data analysis results
- 06-visualization.R: producing figures for the manuscript

You will need to have R and the following packages installed:

- bSims: synthetic simulations
- detect: parameter estimation
- qs: fast read/write for binary R data sets
- rconfig: easily configure R scripts at the command line
- remotes: install packages from non-CRAN repositories
- extraDistr: implementing the triangular distribution that SQPAD uses

```
install.packages(
  pkgs = c('bSims', 'detect', 'qs', 'rconfig', 'mefa4',
    'arrow', 'dplyr', 'ggplot2', 'extraDistr'))
```

If you are running the scripts on a remote server, you might want to set up your R environment on an Ubuntu Linux system following instructions in the `README.md` file.

```
library(ggplot2)
library(dplyr)
library(arrow)
library(mefa4)
library(detect)
library(extraDistr)
```

2 SQPAD examples

2.1 Simple simulated data

First, we showcase the use of the functions in `00-functions-sqpad.R`.

Simulate data under the SQPAD model:

```
n <- 200
x <- rnorm(n)
D <- exp(-2 + 0.5 * x)
phi <- 0.25
tau <- 1
dur <- sample(1:10, n, replace=TRUE)
dis <- sample(seq(0.5, 2, 0.25), n, replace=TRUE)
A <- dis^2 * pi
p <- 1 - exp(-dur * phi * exp(-(dis*0.67)^2/tau^2))
N <- rpois(n, D*A)
Y <- rbinom(n, N, p)

df <- data.frame(x = x, y = Y)
```

Fit the SQPAD model:

```
m <- sqpad(y ~ x | 1, data = df, dis = dis, dur = dur)

print(m)
```

```
##
## Call:
## sqpad(formula = y ~ x | 1, data = df, dis = dis, dur = dur)
##
## Single-bin QPAD model with dependence
## Full Maximum Likelihood estimates
##
## Coefficients:
##      log.D_(Intercept)      log.D_x  log.delta_(Intercept)
##             -2.68663             0.45934             -0.68635
##             log.phi
##             -0.02082
```

```
summary(m)
```

```
##
## Call:
## sqpad(formula = y ~ x | 1, data = df, dis = dis, dur = dur)
##
## Single-bin QPAD model with dependence
## Full Maximum Likelihood estimates
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## log.D_(Intercept)    -2.68663    0.40597  -6.618 3.65e-11 ***
## log.D_x              0.45934    0.15311   3.000  0.0027 **
## log.delta_(Intercept) -0.68635    1.09114  -0.629  0.5293
## log.phi              -0.02082    0.25929  -0.080  0.9360
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-likelihood: -96.07
## BIC = 213.3
```

Methods to extract coefficients, sample size, variance-covariance matrix, confidence intervals:

```
coef(m)
```

```
##      log.D_(Intercept)      log.D_x log.delta_(Intercept)
##      -2.68662989          0.45933681      -0.68635428
##      log.phi
##      -0.02082131
```

```
nobs(m)
```

```
## [1] 200
```

```
vcov(m)
```

```
##              log.D_(Intercept)      log.D_x log.delta_(Intercept)
## log.D_(Intercept)      0.16481548 -0.015630406      -0.33524870
## log.D_x                -0.01563041  0.023441652      0.01865263
## log.delta_(Intercept)  -0.33524870  0.018652627      1.19058680
## log.phi                0.03627967 -0.002516642      -0.23529450
##              log.phi
## log.D_(Intercept)      0.036279673
## log.D_x                -0.002516642
## log.delta_(Intercept) -0.235294504
## log.phi                0.067233273
```

```
confint(m)
```

```
##              2.5 %      97.5 %
## log.D_(Intercept)  -3.4823257 -1.8909340
## log.D_x           0.1592533  0.7594204
## log.delta_(Intercept) -2.8249497  1.4522411
## log.phi           -0.5290278  0.4873852
```

Methods to extract log-likelihood and related information criteria:

```
logLik(m)
```

```
## 'log Lik.' -96.07106 (df=4)
```

```
AIC(m)
```

```
## [1] 200.1421
```

```
BIC(m)
```

```
## [1] 213.3354
```

Methods to get fitted values and predict for new data:

```
summary(fitted(m))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02358 0.05110 0.06630 0.07485 0.08843 0.30183
```

```
summary(predict(m))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -3.747 -2.974 -2.714 -2.691 -2.426 -1.198
```

```
summary(predict(m, type = "response"))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02358 0.05110 0.06630 0.07485 0.08843 0.30183
```

```
predict(m, newdata = df[1:10,], type = "link")
```

```
##           1           2           3           4           5           6           7           8
## -2.944077 -2.792359 -1.970658 -2.654243 -2.627243 -1.898837 -2.474914 -3.267719
##           9          10
## -3.002127 -2.891339
```

```
predict(m, newdata = df[1:10,], type = "response")
```

```
##           1           2           3           4           5           6           7
## 0.05265063 0.06127650 0.13936516 0.07035209 0.07227744 0.14974261 0.08417022
##           8           9          10
## 0.03809322 0.04968130 0.05550185
```

Model selection using different models (m is the true model):

```
m0 <- sqpad(y ~ 1 | 1, data = df, dis = dis, dur = dur)
```

```
m1 <- sqpad(y ~ 1 | x, data = df, dis = dis, dur = dur)
```

```
m2 <- sqpad(y ~ x | x, data = df, dis = dis, dur = dur)
```

```
AIC(m, m0, m1, m2)
```

```
##      df      AIC
## m      4 200.1421
## m0     3 206.8932
## m1     4 201.3077
## m2     5 202.0939
```

```
BIC(m, m0, m1, m2)
```

```
##      df      BIC
## m      4 213.3354
## m0     3 216.7882
```

```
## m1  4 214.5010
## m2  5 218.5855
```

2.2 Using the convolution model

Simulated data set and estimation:

```
D <- 1
phi <- 0.25
tau <- 1.2
n <- 200

dur <- sample(1:10, n, replace=TRUE)
dis <- sample(seq(0.25, 2, 0.25), n, replace=TRUE)
A <- dis^2 * pi
N <- rpois(n, D*A)
dislist <- lapply(seq_len(n), function(i) {
  dij <- extraDistr::rtriang(N[i], 0, dis[i], dis[i])
  # pij is based on time of 1st detection and distance at 1st detection
  pij <- 1 - exp(-dur[i] * phi * exp(-(dij)^2/tau^2))
  Yij <- rbinom(length(pij), 1, pij)
  dij[Yij > 0]
})
Y <- sapply(dis, length)

m4 <- sqpad.fit(Y = Y, dis = dis, dur = dur, dislist = dislist,
  type = "conv", det = "joint", Nmax = 20)
cbind(parameters = c(D = D, phi = phi, tau = tau), estimates = exp(m4$coef))

##      parameters estimates
## D          1.00 1.2369595
## phi         0.25 0.1689902
## tau         1.20 1.2828854
```

2.3 Field data example

This example uses Ovenbird as an example from the field data section and demonstrates how to use multiple-bin data using the composite likelihood approach to estimating model parameters.

Data manipulation:

```
x <- detect::paired
x$UniqueID <- as.factor(x$UniqueID)
x <- droplevels(x[x$SurveyType == "HUM" & x$Visit == 1 & !is.na(x$Latitude),])
ID <- levels(x$UniqueID)

spp <- "OVEN" # species ID

id <- ID # site IDs to consider

# when doing bootstrap, resample site IDs with replacement as
# id <- sample(ID, length(ID), replace=TRUE)
```

Naive estimator:

```
d <- nonDuplicated(x, UniqueID, TRUE)
n <- nrow(d)
```

```

X <- model.matrix(~scale(Latitude), d)
rn <- rownames(d)

yydis <- as.matrix(Xtab(Count ~ UniqueID + DISTANCE,
  x[x$SPECIES == spp,]))
yydis <- yydis[rn,c("0-49 m", "50-100 m")]
Y <- rowSums(yydis)
Area <- rep(1^2 * pi, n) # 100m radius

rn <- id
mod_naive <- glm(count ~ lat, data.frame(count = Y[rn], lat=X[rn,2]),
  offset=log(Area), family=poisson)

unname(mean(exp(X %*% coef(mod_naive))))

## [1] 0.3319129

QPAD estimator:

d <- nonDuplicated(x, UniqueID, TRUE)
n <- nrow(d)
X <- model.matrix(~scale(Latitude), d)
Z <- model.matrix(~TSSR+I(TSSR^2), d)
rn <- id

# distance
yydis <- as.matrix(Xtab(Count ~ UniqueID + DISTANCE,
  x[x$SPECIES == spp,]))[rn,c("0-49 m", "50-100 m")]
dddis <- matrix(c(0.5, 1), nrow(yydis), 2, byrow=TRUE)
mdis <- cmulti.fit(Y=yydis, D=dddis, type = "dis")

# time
yydur <- as.matrix(Xtab(Count ~ UniqueID + Interval,
  x[x$SPECIES == spp & x$DISTANCE %in% c("0-49 m", "50-100 m"),]))[rn,
  c("0-3 min", "3-5 min", "5-10 min")]
dddur <- matrix(c(3, 5, 10), nrow(yydur), 3, byrow=TRUE)
mdur <- cmulti.fit(Y=yydur, D=dddur, X=Z, type = "rem")

# QPAD
Y <- rowSums(yydis)
TAU <- exp(coef(mdis))
PHI <- exp(Z %*% coef(mdur))
Area <- rep(1^2 * pi, n) # 100m radius
p <- 1 - exp(-PHI * 10)
r <- 1
q <- TAU^2 * (1 - exp(-r^2/TAU^2))/r^2
Off <- log(Area * p * q)

mod_qpad <- glm(Y ~ lat, data.frame(lat=X[rn,2]), offset = Off, family=poisson)

unname(mean(exp(X %*% coef(mod_qpad))))

## [1] 0.6306315

SQPAD estimator:

```

```

x$UniqueID <- as.character(x$UniqueID)
xx <- do.call(rbind, lapply(1:length(id), function(i) {
  z <- x[x$UniqueID==id[i],]
  z$UniqueID <- paste0(z$UniqueID, "_", i)
  z
}))
xx$UniqueID <- as.factor(xx$UniqueID)
xt <- Xtab(Count ~ UniqueID + Interval + DISTANCE, xx[xx$SPECIES == spp,])
d <- nonDuplicated(xx, UniqueID, TRUE)
n <- nrow(d)

y11 <- xt[["0-49 m"]][,c("0-3 min")]
y12 <- rowSums(xt[["0-49 m"]][,c("0-3 min", "3-5 min")])
y13 <- rowSums(xt[["0-49 m"]][,c("0-3 min", "3-5 min", "5-10 min")])
y21 <- y11 + xt[["50-100 m"]][,c("0-3 min")]
y22 <- y12 + rowSums(xt[["50-100 m"]][,c("0-3 min", "3-5 min")])
y23 <- y13 + rowSums(xt[["50-100 m"]][,c("0-3 min", "3-5 min", "5-10 min")])
y31 <- y11 + y21 + xt[[">100 m"]][,c("0-3 min")]
y32 <- y12 + y22 + rowSums(xt[[">100 m"]][,c("0-3 min", "3-5 min")])
y33 <- y13 + y23 + rowSums(xt[[">100 m"]][,c("0-3 min", "3-5 min", "5-10 min")])

dis <- rep(c(0.5, 1), each = 3 * n)
dur <- c(rep(c(3, 5, 10), each = n), rep(c(3, 5, 10), each = n))
pid <- rep(1:n, 2*3)
y <- c(y11, y12, y13, y21, y22, y23)

X <- model.matrix(~scale(Latitude), d)[pid,]
Z <- model.matrix(~TSSR+I(TSSR^2), d)[pid,]

mod_sqpad_count <- sqpad.fit(Y=y, dis=dis, dur=dur, X=X, Z=Z,
  type = "full", det = "joint", K = NULL, hessian = FALSE)

mod_sqpad_occ <- sqpad.fit(Y=y, dis=dis, dur=dur, X=X, Z=Z,
  type = "full", det = "joint", K = 1, hessian = FALSE)

unname(mean(exp(X %*% coef(mod_sqpad_count)[1:ncol(X)])))

## [1] 0.4247175

unname(mean(exp(X %*% coef(mod_sqpad_occ)[1:ncol(X)])))

## [1] 0.4004753

```

3 Synthetic simulations

We can easily modify simulation settings using the `02-bsims-simulations.R` R script at the command line. These will produce output files in the `_tmp/bsims` folder.

The following command line arguments can be specified:

- `all`: violate all assumptions (overrides `move`, `disterr`, `doublecount`, and `jointdist`)
- `move`: introduce movement
- `disterr`: introduce distance estimation error
- `doublecount`: introduce double counting
- `jointdist`: introduce joint distribution of time and distance

- **n**: number of independent realizations to simulate
- **D**: value for the population density (individuals/ha)
- **phi**: value for the innate cue rate parameter (1/min)
- **tau**: value for the distance parameter (in 100 m units)
- **dir**: output folder for saving results into

You can vary these arguments. We used the following settings:

```
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.25 --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.25 --jointdist --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.25 --all --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.25 --jointdist --move --disterr --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.25 --jointdist --move --doublecount --n 1000

Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.5 --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.5 --jointdist --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.5 --all --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.5 --jointdist --move --disterr --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 0.5 --phi 0.5 --jointdist --move --doublecount --n 1000

Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.25 --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.25 --jointdist --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.25 --all --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.25 --jointdist --move --disterr --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.25 --jointdist --move --doublecount --n 1000

Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.5 --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.5 --jointdist --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.5 --all --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.5 --jointdist --move --disterr --n 1000
Rscript --vanilla 02-bsims-simulations.R --D 1 --phi 0.5 --jointdist --move --doublecount --n 1000
```

4 Estimating parameters from simulations

The 03-bsims-estimation.R will estimate model parameters for various methods and write results into a file inside the `_tmp/est_conv` folder.

The following command line arguments can be specified:

- **dir**: output folder for saving results into
- **B**: how many replicates to run
- **n**: sample size for each replicate
- **set**: one of the 16 setups we have in the `_tmp/bsims` folder
- **replace**: whether sites (simulation realizations) should be picked with replacement
- **notify**: flag when specified, it sends notifications via <https://docs.ntfy.sh/>

You can vary these arguments. We used the following settings:

```
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 1
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 2
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 3
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 4

Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 5
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 6
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 7
```



```

Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 8

Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 9
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 10
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 11
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 12

Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 13
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 14
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 15
Rscript --vanilla 03-bsims-estimation.R --B 2000 --notify --set 16

```

5 Field data analysis

Run analyses in the 04-field-data.R R script. The id parameter refer to one of the 12 species with sufficient data for analyses:

The following command line arguments can be specified:

- dir: output folder for saving results into
- B: how many nonparametric bootstrap iterations to run
- id: species ID

You can vary these arguments. We used the following settings:

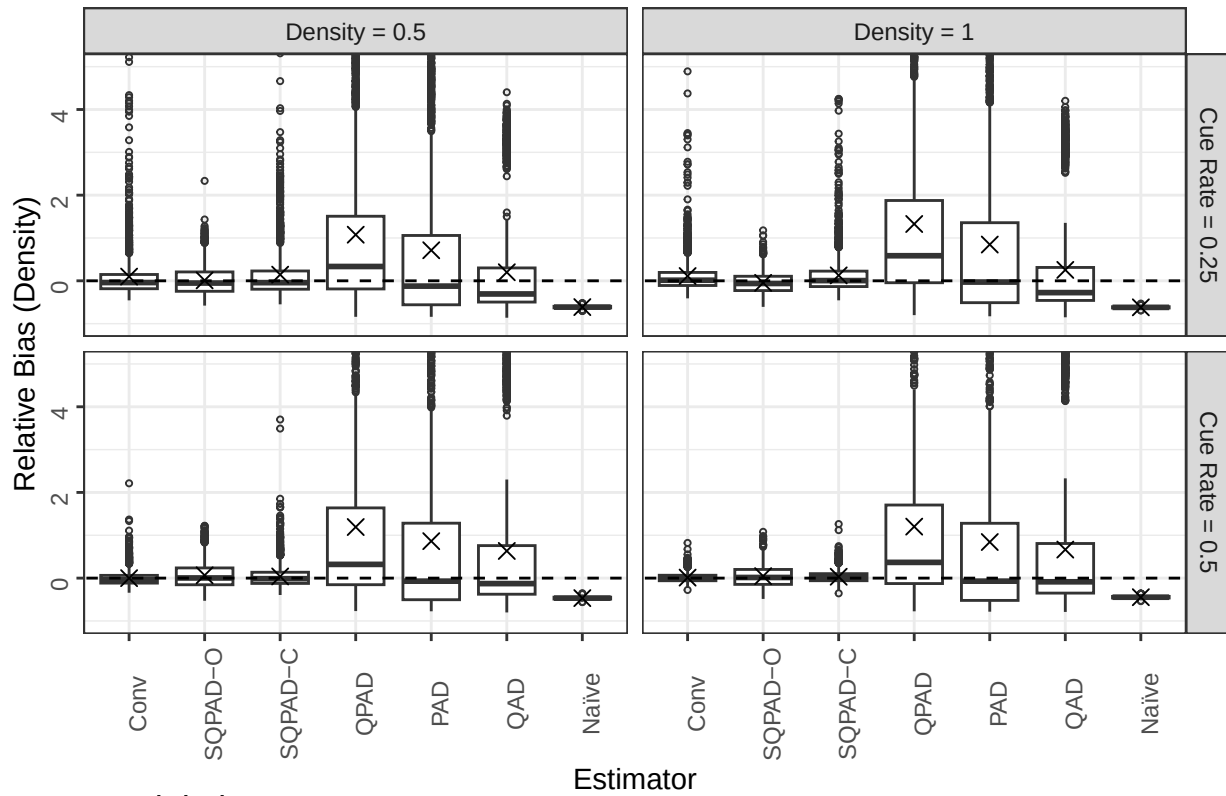
```

Rscript --vanilla 04-field-data.R --id 1
Rscript --vanilla 04-field-data.R --id 2
Rscript --vanilla 04-field-data.R --id 3
Rscript --vanilla 04-field-data.R --id 4
Rscript --vanilla 04-field-data.R --id 5
Rscript --vanilla 04-field-data.R --id 6
Rscript --vanilla 04-field-data.R --id 7
Rscript --vanilla 04-field-data.R --id 8
Rscript --vanilla 04-field-data.R --id 9
Rscript --vanilla 04-field-data.R --id 10
Rscript --vanilla 04-field-data.R --id 11
Rscript --vanilla 04-field-data.R --id 12

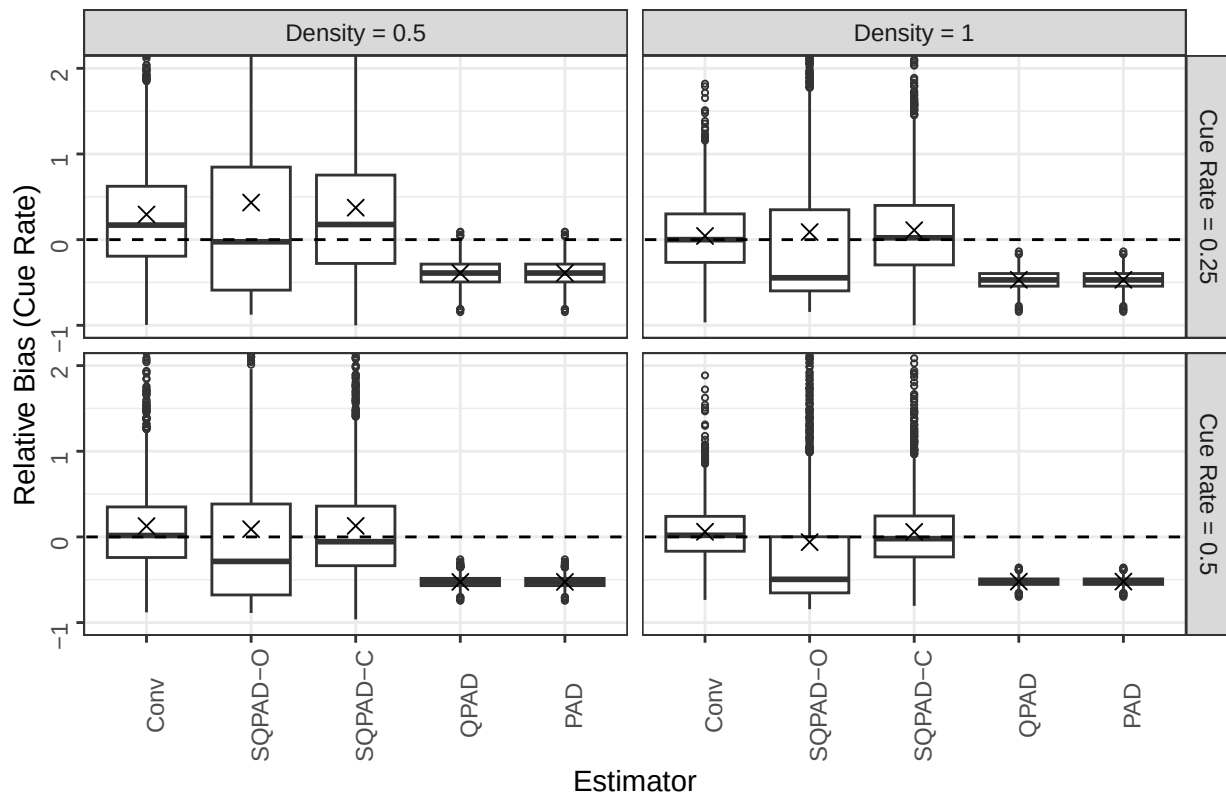
```

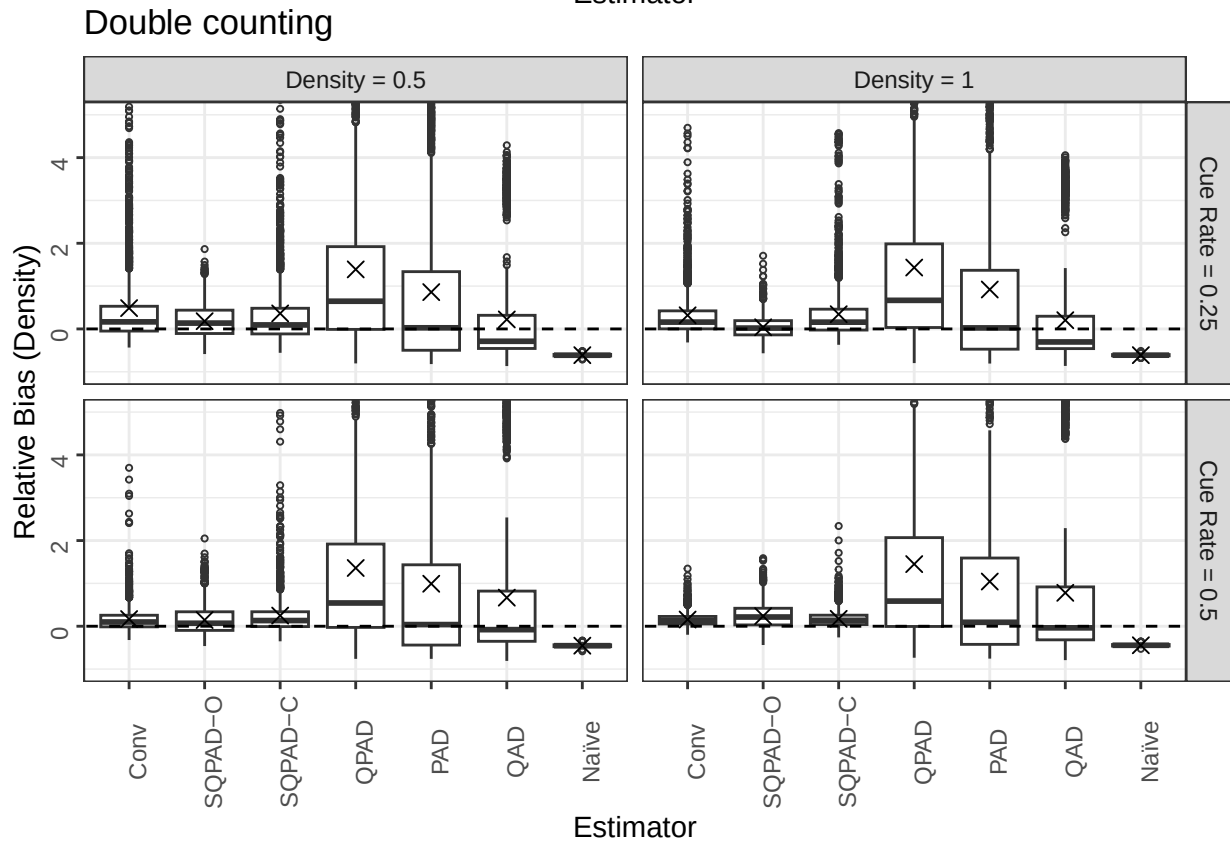
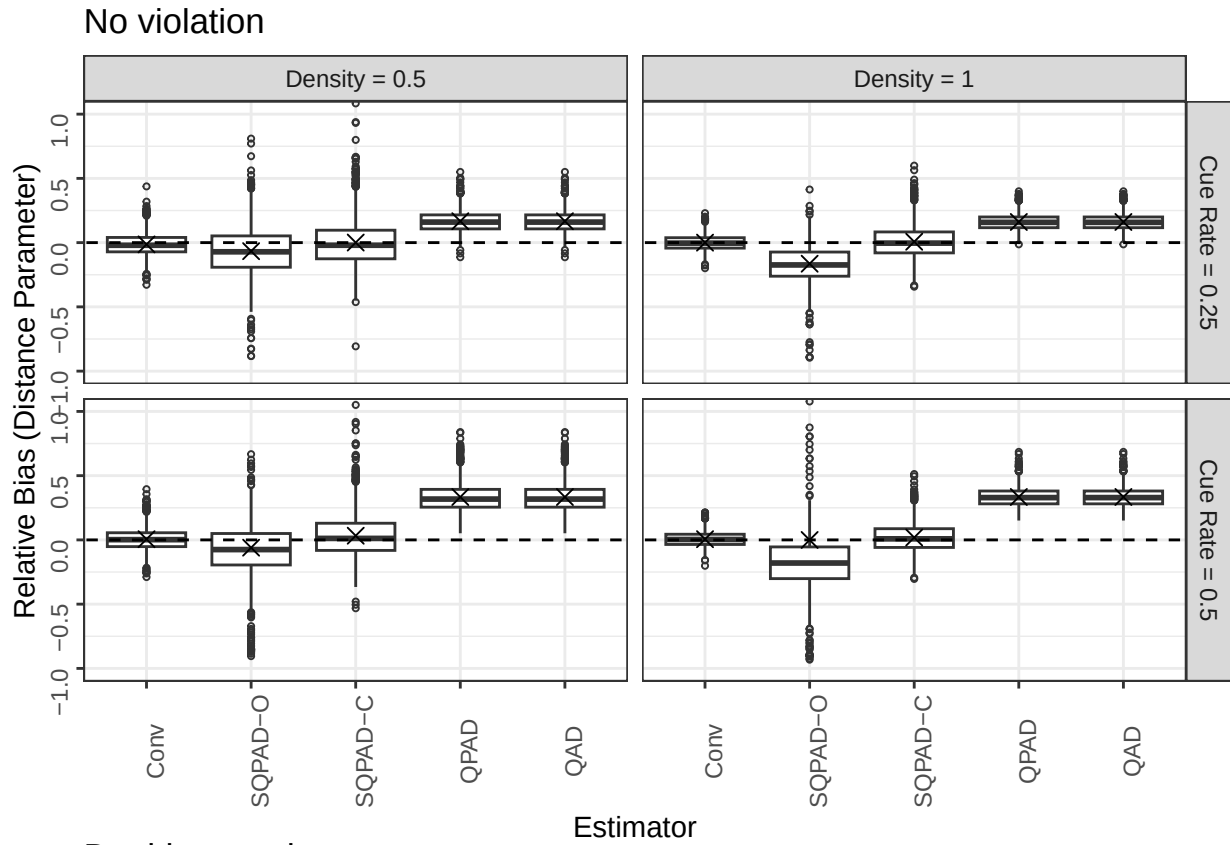
6 Simulation results

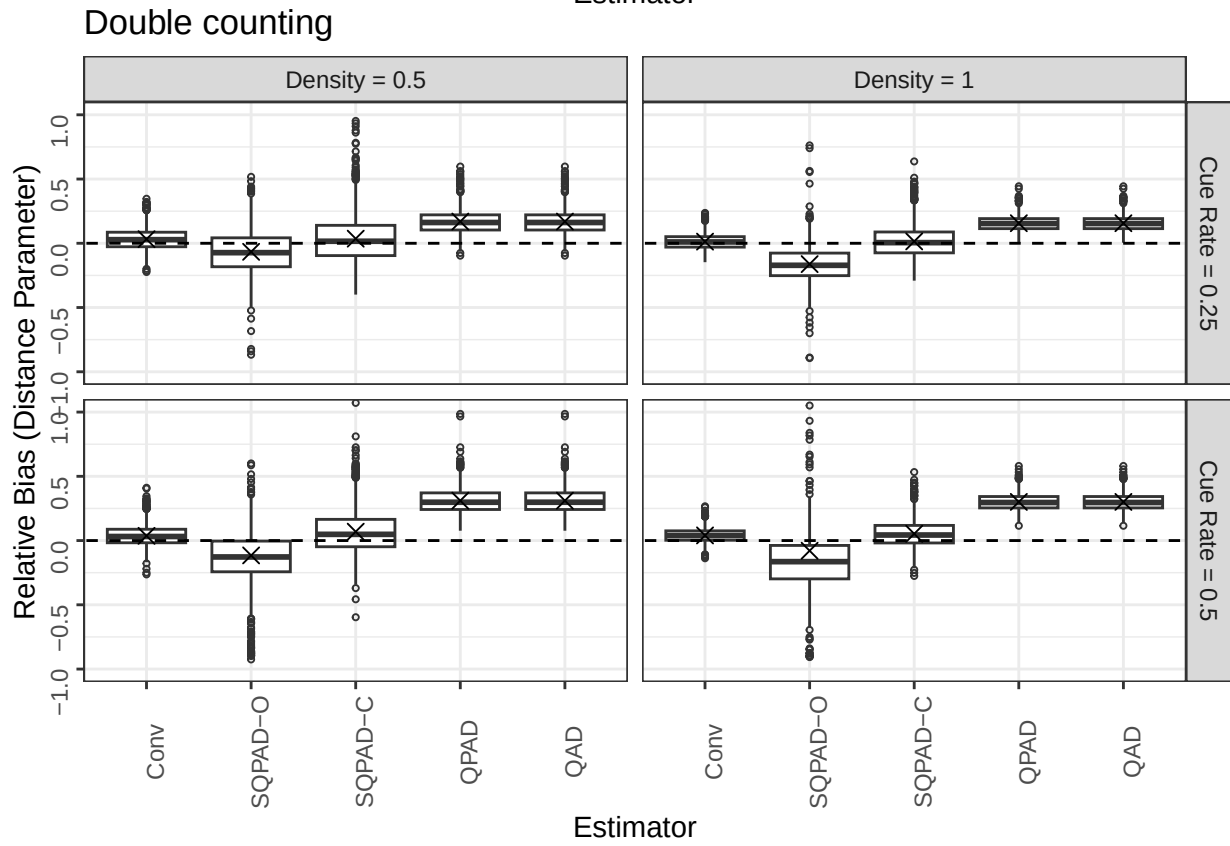
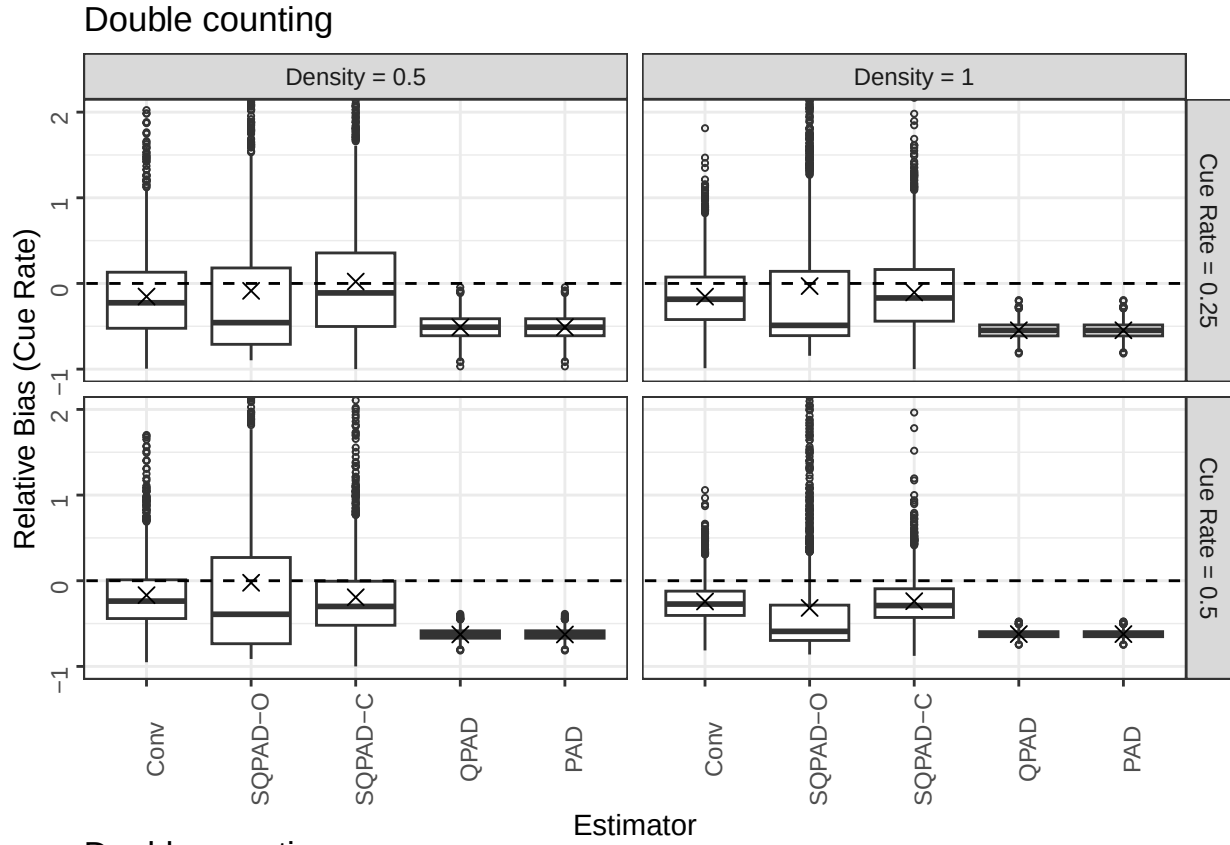
No violation

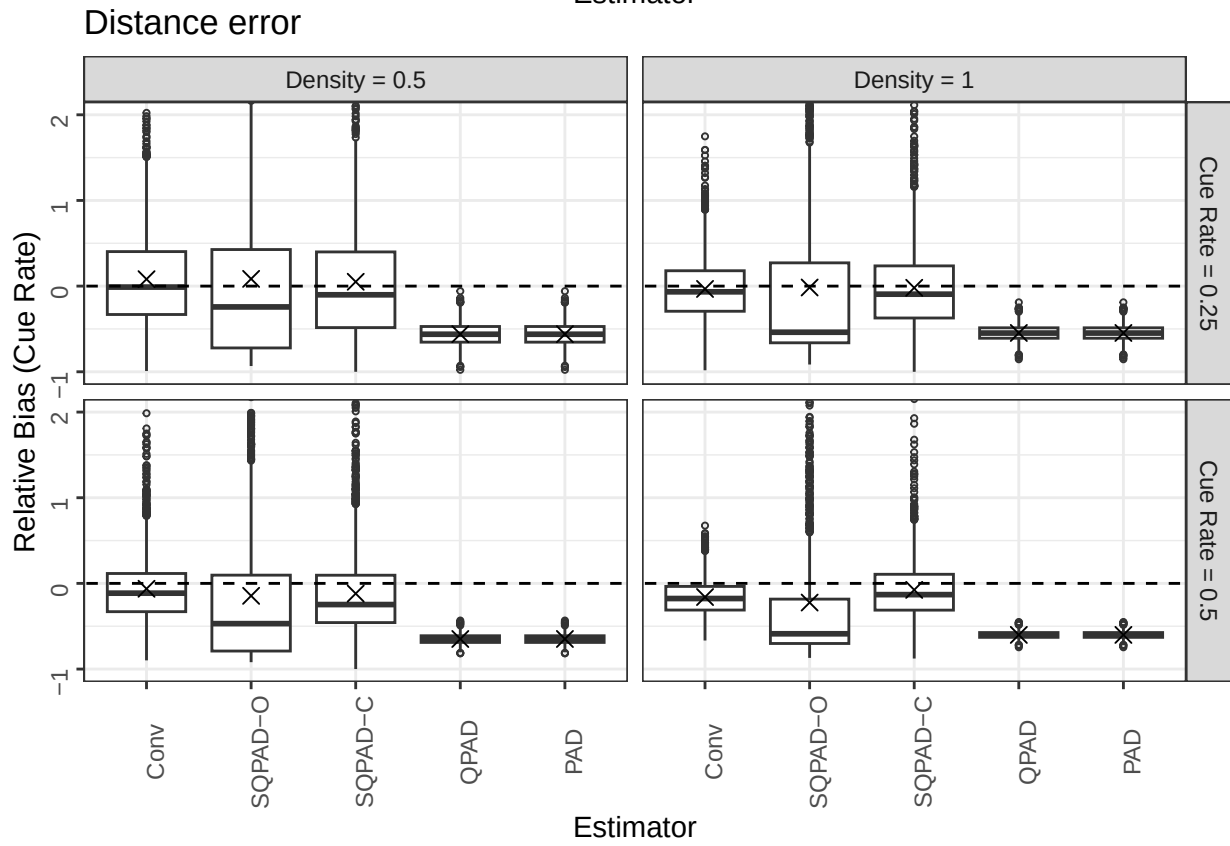
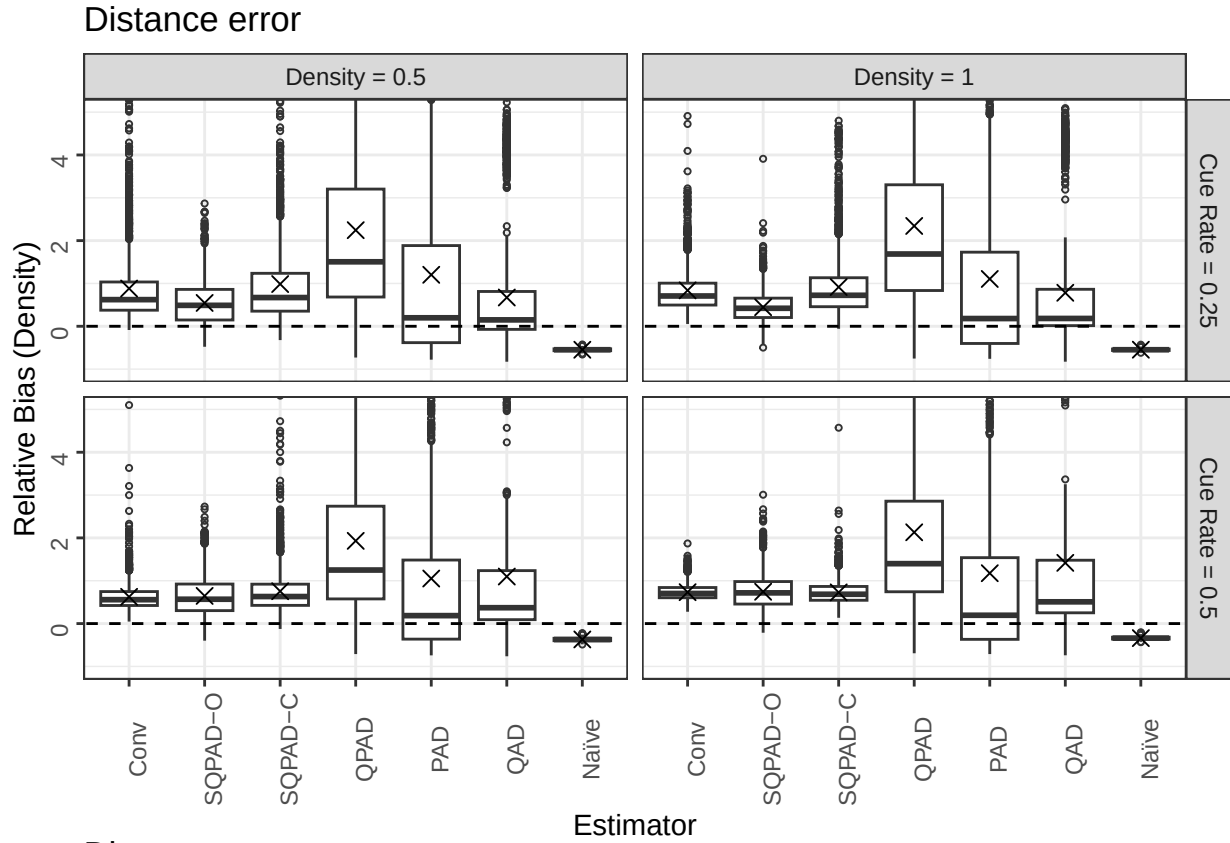


No violation









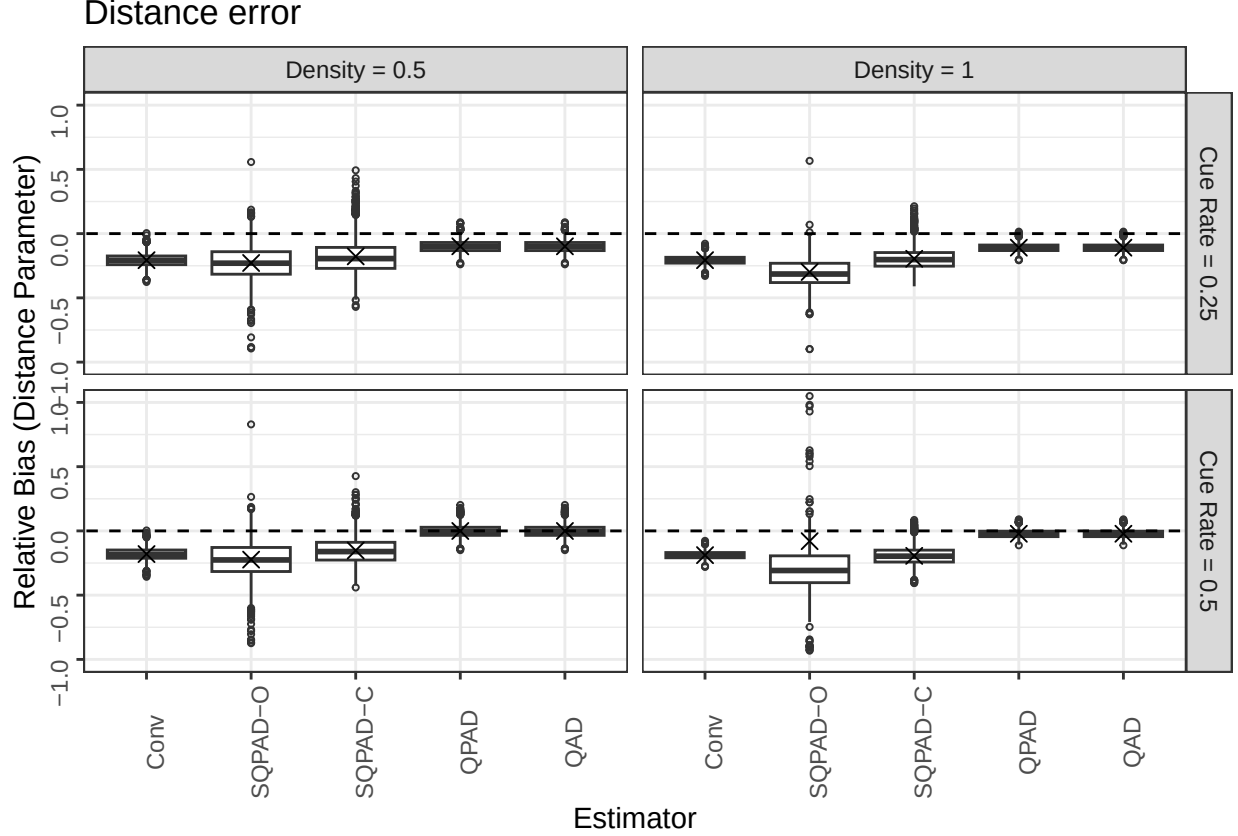


Table 1: Mean and Standard Error for the relative bias for Density,
No violation

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	0.098 [0.614]	-0.001 [0.187]	0.112 [0.402]	0.012 [0.105]
SQPAD-O	0.01 [0.334]	0.065 [0.294]	-0.048 [0.247]	0.046 [0.247]
SQPAD-C	0.144 [0.741]	0.039 [0.275]	0.127 [0.485]	0.031 [0.144]
QPAD	1.077 [1.956]	1.192 [2.116]	1.327 [2.013]	1.202 [2.068]
PAD	0.713 [1.972]	0.863 [2.142]	0.847 [2.082]	0.842 [2.103]
QAD	0.2 [1.216]	0.634 [1.706]	0.255 [1.203]	0.666 [1.69]
Naïve	-0.614 [0.028]	-0.467 [0.032]	-0.619 [0.023]	-0.446 [0.027]

Table 2: Mean and Standard Error for the relative bias for Density,
Double counting

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	0.487 [1.075]	0.167 [0.337]	0.319 [0.605]	0.156 [0.155]
SQPAD-O	0.181 [0.371]	0.145 [0.336]	0.039 [0.26]	0.236 [0.279]
SQPAD-C	0.362 [0.916]	0.248 [0.528]	0.344 [0.674]	0.173 [0.209]
QPAD	1.39 [2.064]	1.359 [2.141]	1.434 [2.105]	1.45 [2.187]
PAD	0.86 [2.051]	0.992 [2.192]	0.92 [2.129]	1.043 [2.235]
QAD	0.222 [1.178]	0.667 [1.708]	0.204 [1.195]	0.78 [1.776]
Naïve	-0.613 [0.029]	-0.454 [0.034]	-0.612 [0.022]	-0.445 [0.027]

Table 3: Mean and Standard Error for the relative bias for Density,
Distance error

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	0.882 [0.898]	0.624 [0.331]	0.839 [0.582]	0.732 [0.19]
SQPAD-O	0.545 [0.51]	0.646 [0.467]	0.445 [0.371]	0.743 [0.408]
SQPAD-C	0.986 [1.081]	0.755 [0.612]	0.916 [0.74]	0.728 [0.286]
QPAD	2.244 [2.21]	1.931 [2.042]	2.343 [2.078]	2.131 [2.069]
PAD	1.202 [2.305]	1.052 [2.148]	1.105 [2.199]	1.176 [2.356]
QAD	0.672 [1.392]	1.098 [1.887]	0.782 [1.388]	1.418 [2.031]
Naïve	-0.548 [0.033]	-0.371 [0.04]	-0.547 [0.027]	-0.34 [0.033]

Table 4: Mean and Standard Error for the relative bias for Cue
Rate, No violation

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	0.294 [0.772]	0.126 [0.613]	0.043 [0.434]	0.06 [0.326]
SQPAD-O	0.433 [1.454]	0.09 [1.181]	0.089 [1.17]	-0.061 [1.189]
SQPAD-C	0.373 [1.001]	0.129 [0.773]	0.111 [0.576]	0.059 [0.443]
QPAD	-0.389 [0.15]	-0.525 [0.066]	-0.469 [0.106]	-0.522 [0.048]
PAD	-0.389 [0.15]	-0.525 [0.066]	-0.469 [0.106]	-0.522 [0.048]

Table 5: Mean and Standard Error for the relative bias for Cue
Rate, Double counting

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	-0.153 [0.542]	-0.168 [0.422]	-0.154 [0.38]	-0.242 [0.232]
SQPAD-O	-0.086 [1.012]	-0.024 [1.143]	-0.031 [1.071]	-0.317 [0.799]
SQPAD-C	0.022 [0.776]	-0.193 [0.529]	-0.105 [0.483]	-0.239 [0.285]
QPAD	-0.51 [0.143]	-0.627 [0.062]	-0.546 [0.097]	-0.623 [0.043]
PAD	-0.51 [0.143]	-0.627 [0.062]	-0.546 [0.097]	-0.623 [0.043]

Table 6: Mean and Standard Error for the relative bias for Cue
Rate, Distance error

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	0.081 [0.631]	-0.063 [0.425]	-0.033 [0.373]	-0.159 [0.209]
SQPAD-O	0.085 [1.131]	-0.145 [1.033]	-0.016 [1.14]	-0.224 [1.012]
SQPAD-C	0.05 [0.841]	-0.119 [0.535]	-0.02 [0.506]	-0.076 [0.343]
QPAD	-0.561 [0.133]	-0.649 [0.058]	-0.547 [0.091]	-0.601 [0.041]
PAD	-0.561 [0.133]	-0.649 [0.058]	-0.547 [0.091]	-0.601 [0.041]

Table 7: Mean and Standard Error for the relative bias for Distance
Parameter, No violation

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	-0.014 [0.085]	0.006 [0.085]	0 [0.061]	0.006 [0.059]
SQPAD-O	-0.066 [0.209]	-0.06 [0.413]	-0.164 [0.23]	-0.001 [1.064]
SQPAD-C	0.001 [0.184]	0.033 [0.168]	0.007 [0.125]	0.019 [0.113]
QPAD	0.167 [0.088]	0.334 [0.116]	0.161 [0.064]	0.334 [0.075]
QAD	0.167 [0.088]	0.334 [0.116]	0.161 [0.064]	0.334 [0.075]

Table 8: Mean and Standard Error for the relative bias for Distance
Parameter, Double counting

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	0.032 [0.086]	0.037 [0.082]	0.013 [0.059]	0.04 [0.055]
SQPAD-O	-0.066 [0.175]	-0.116 [0.368]	-0.162 [0.164]	-0.079 [0.701]
SQPAD-C	0.036 [0.192]	0.069 [0.172]	0.015 [0.125]	0.053 [0.106]
QPAD	0.169 [0.092]	0.311 [0.098]	0.156 [0.06]	0.301 [0.066]
QAD	0.169 [0.092]	0.311 [0.098]	0.156 [0.06]	0.301 [0.066]

Table 9: Mean and Standard Error for the relative bias for Distance
Parameter, Distance error

	D=0.5, delta=0.25	D=0.5, delta=0.5	D=1, delta=0.25	D=1, delta=0.5
Conv	-0.207 [0.052]	-0.18 [0.05]	-0.207 [0.034]	-0.189 [0.03]
SQPAD-O	-0.228 [0.142]	-0.223 [0.153]	-0.299 [0.275]	-0.08 [1.124]
SQPAD-C	-0.177 [0.136]	-0.152 [0.109]	-0.196 [0.085]	-0.194 [0.072]
QPAD	-0.099 [0.047]	-0.001 [0.051]	-0.109 [0.032]	-0.022 [0.032]
QAD	-0.099 [0.047]	-0.001 [0.051]	-0.109 [0.032]	-0.022 [0.032]

6.1 Mean distance vs. average detection probability

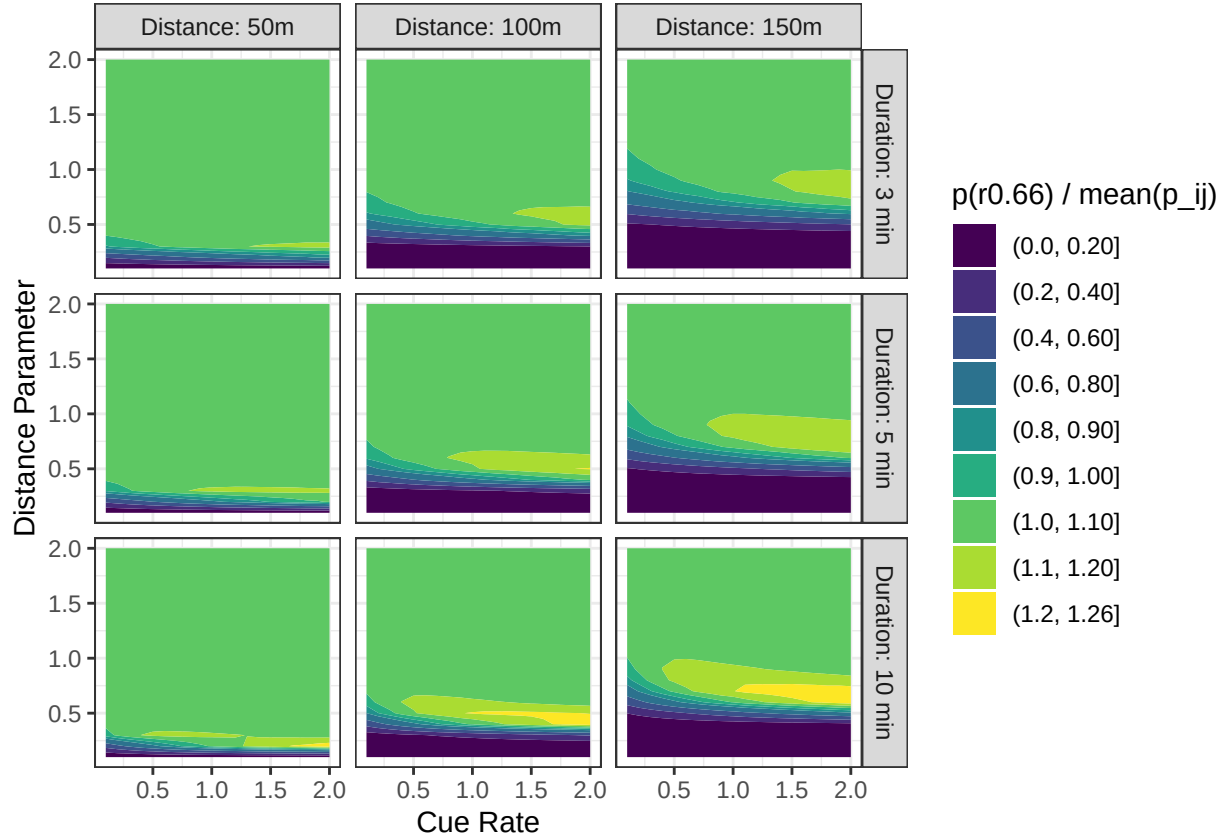
The `montecarlo` argument of the `sqpad` function can be set to:

- `TRUE`: use Monte Carlo average probability,
- `FALSE`: use the mead distance to calculate an approximate probability.

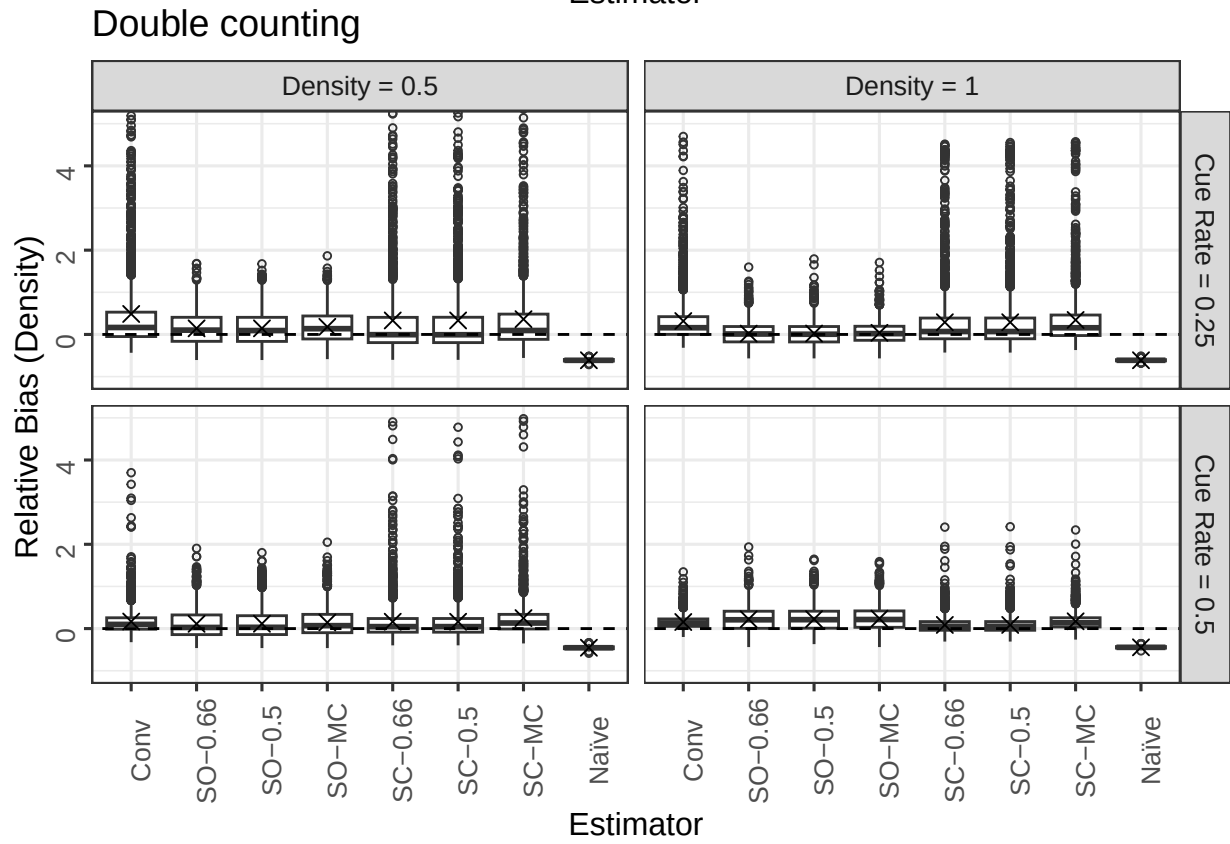
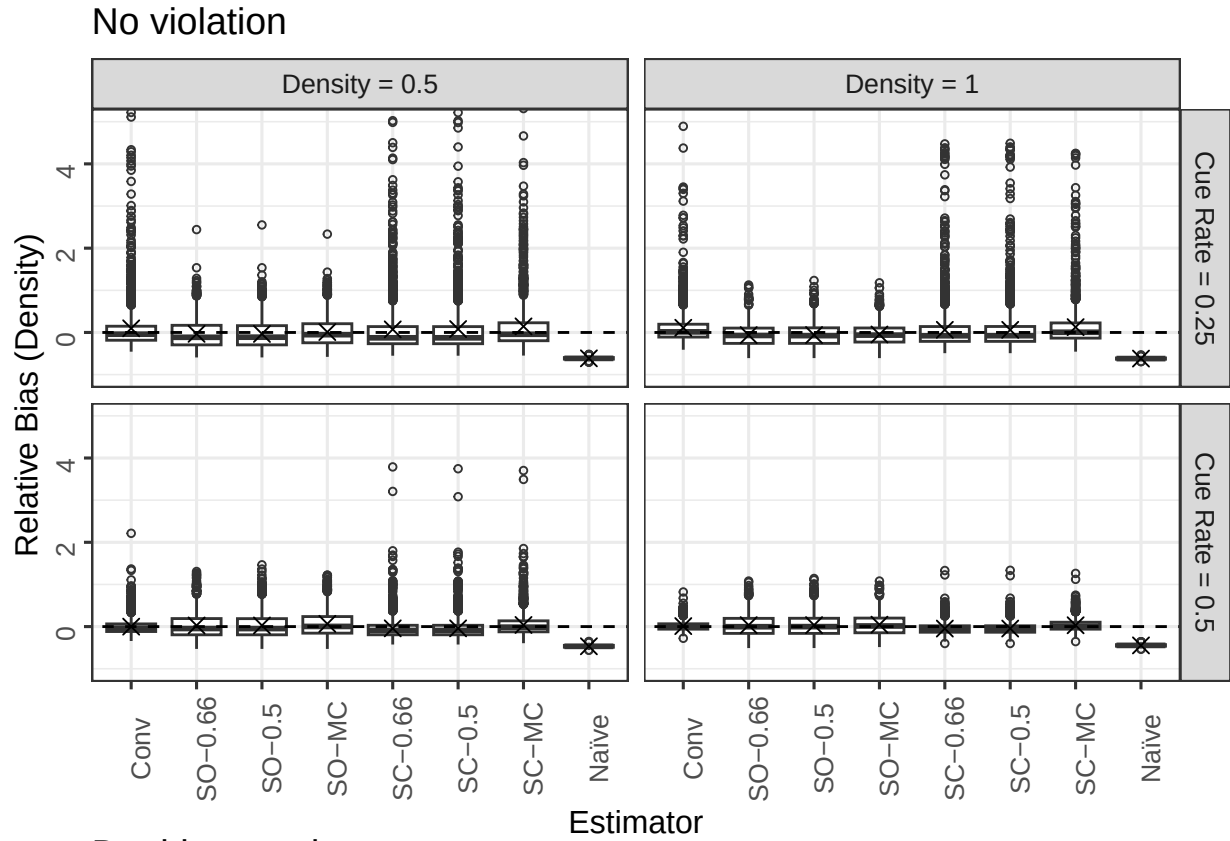
The following contour plot shows the detection probabilities using the two approaches based on different settings for:

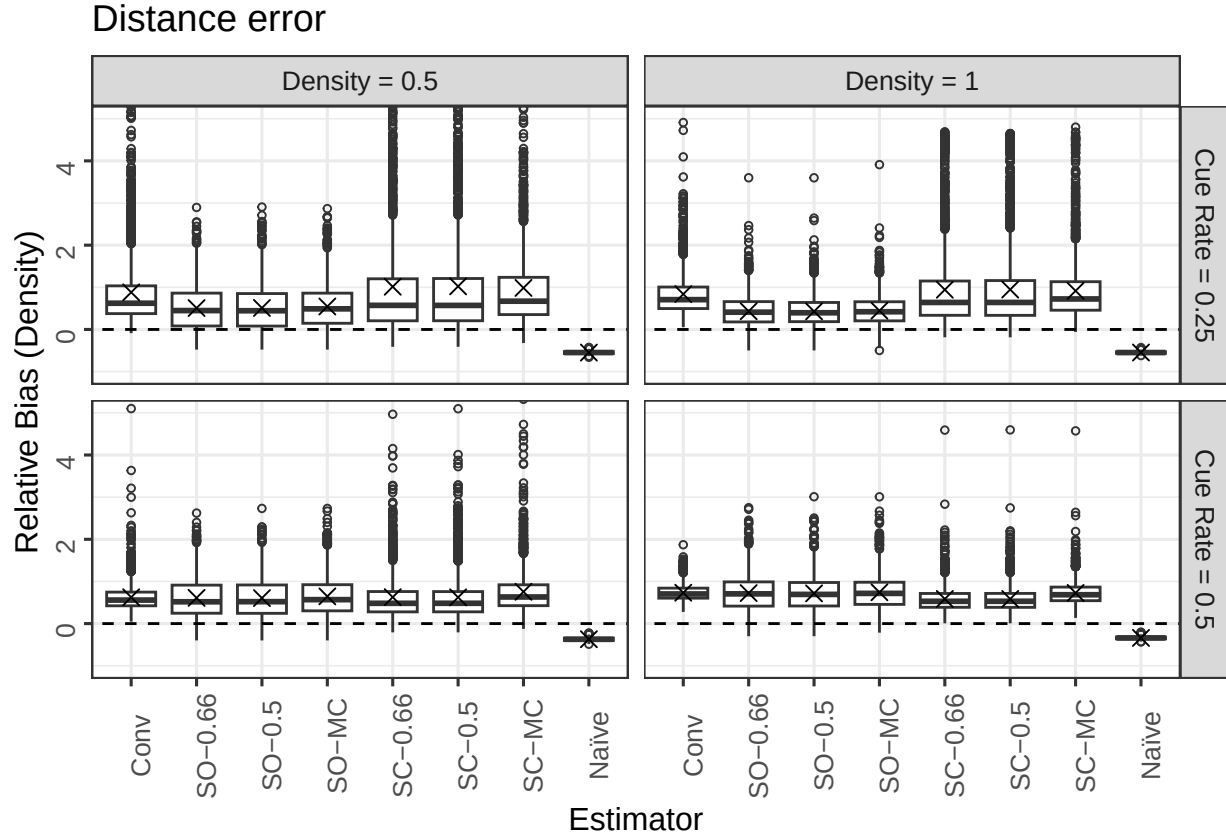
- Cue rate,
- Distance parameter,
- Duration (in minutes),
- Distance (in meters).

The mean distance based probability (`p(r0.66)`) can be lower than the Monte Carlo based mean probability (`mean(p_ij)`) for small values of the detection parameter and large distances:



We also checked how these approaches compare when estimating population density. We compared the Monte Carlo approach with the $p(r0.66)$ and $p(r0.5)$ approximations. These all provided very similar results in terms of the distribution of the density (D) estimates. We compared the SQPAD occupancy (SO) and SQPAD count model (SC) to the Convolution and Naive estimators:





7 Field data results

Table 10: Species used in field data analysis

Species Code	Species Name
CHSP	Chipping Sparrow
DEJU	Dark-eyed Junco
HETH	Hermit Thrush
MAWA	Magnolia Warbler
OVEN	Ovenbird
RCKI	Ruby-crowned Kinglet
REVI	Red-eyed Vireo
SWTH	Swainson's Thrush
TEWA	Tennessee Warbler
WIWR	Winter Wren
WTSP	White-throated Sparrow

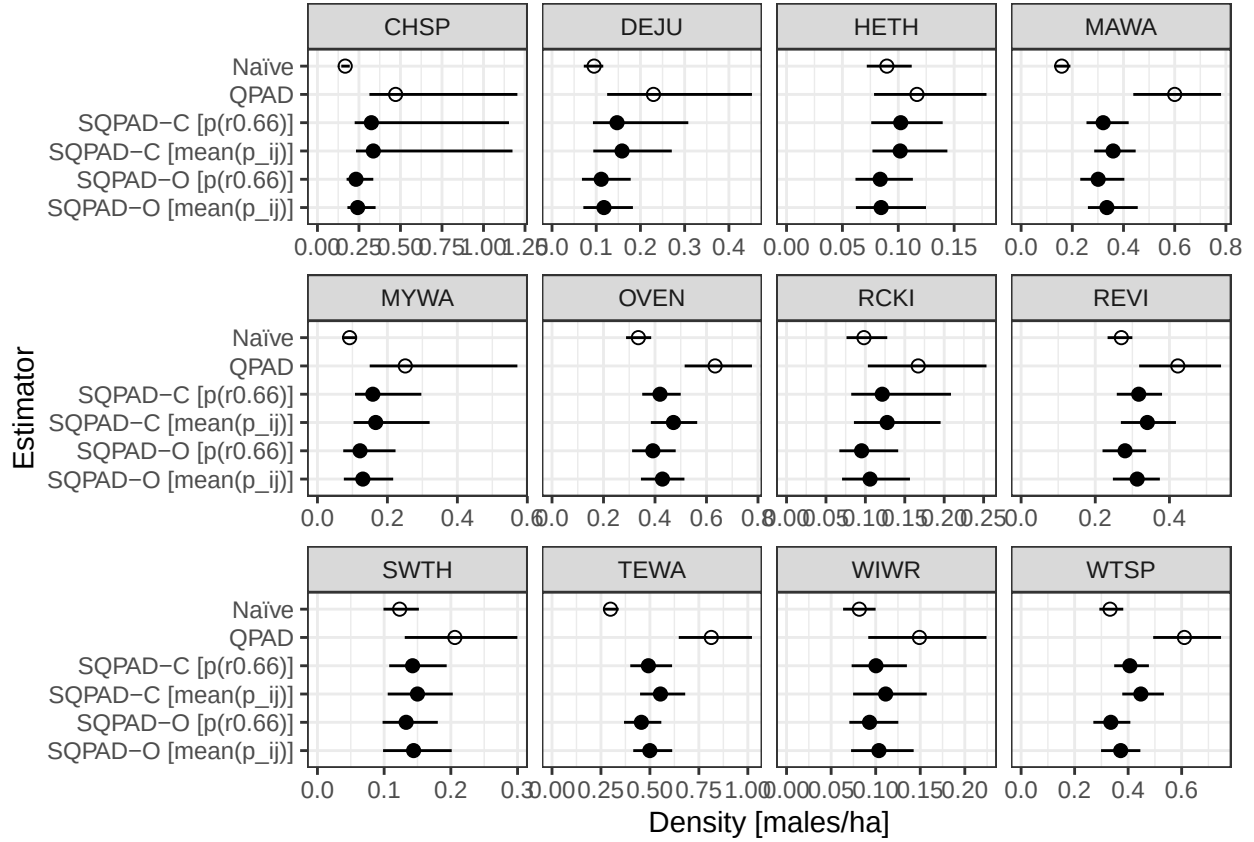


Table 11: Density estimates (median and 95% confidence limits) for field data

species	method	median	lower	upper
CHSP	SQPAD-O [mean(p _{ij})]	0.242	0.181	0.350
CHSP	SQPAD-O [p(r0.66)]	0.232	0.176	0.336
CHSP	SQPAD-C [mean(p _{ij})]	0.336	0.232	1.175
CHSP	SQPAD-C [p(r0.66)]	0.325	0.224	1.153
CHSP	QPAD	0.471	0.313	1.204
CHSP	Naïve	0.167	0.144	0.194
DEJU	SQPAD-O [mean(p _{ij})]	0.118	0.071	0.183
DEJU	SQPAD-O [p(r0.66)]	0.111	0.068	0.178
DEJU	SQPAD-C [mean(p _{ij})]	0.158	0.093	0.271
DEJU	SQPAD-C [p(r0.66)]	0.147	0.093	0.308
DEJU	QPAD	0.229	0.125	0.452
DEJU	Naïve	0.095	0.072	0.116
HETH	SQPAD-O [mean(p _{ij})]	0.085	0.062	0.125
HETH	SQPAD-O [p(r0.66)]	0.084	0.062	0.113
HETH	SQPAD-C [mean(p _{ij})]	0.102	0.077	0.144
HETH	SQPAD-C [p(r0.66)]	0.102	0.076	0.140
HETH	QPAD	0.117	0.078	0.179
HETH	Naïve	0.090	0.072	0.112
MAWA	SQPAD-O [mean(p _{ij})]	0.335	0.261	0.456
MAWA	SQPAD-O [p(r0.66)]	0.301	0.231	0.403
MAWA	SQPAD-C [mean(p _{ij})]	0.359	0.286	0.448
MAWA	SQPAD-C [p(r0.66)]	0.320	0.255	0.420

species	method	median	lower	upper
MAWA	QPAD	0.600	0.438	0.781
MAWA	Naïve	0.159	0.132	0.193
MYWA	SQPAD-O [mean(p_ij)]	0.130	0.075	0.217
MYWA	SQPAD-O [p(r0.66)]	0.122	0.074	0.223
MYWA	SQPAD-C [mean(p_ij)]	0.167	0.103	0.321
MYWA	SQPAD-C [p(r0.66)]	0.158	0.107	0.297
MYWA	QPAD	0.251	0.149	0.572
MYWA	Naïve	0.092	0.074	0.113
OVEN	SQPAD-O [mean(p_ij)]	0.429	0.345	0.514
OVEN	SQPAD-O [p(r0.66)]	0.392	0.311	0.480
OVEN	SQPAD-C [mean(p_ij)]	0.472	0.384	0.564
OVEN	SQPAD-C [p(r0.66)]	0.419	0.350	0.500
OVEN	QPAD	0.634	0.515	0.777
OVEN	Naïve	0.335	0.288	0.385
RCKI	SQPAD-O [mean(p_ij)]	0.106	0.070	0.157
RCKI	SQPAD-O [p(r0.66)]	0.095	0.067	0.142
RCKI	SQPAD-C [mean(p_ij)]	0.128	0.085	0.195
RCKI	SQPAD-C [p(r0.66)]	0.121	0.082	0.209
RCKI	QPAD	0.167	0.103	0.254
RCKI	Naïve	0.098	0.076	0.128
REVI	SQPAD-O [mean(p_ij)]	0.313	0.247	0.374
REVI	SQPAD-O [p(r0.66)]	0.280	0.220	0.337
REVI	SQPAD-C [mean(p_ij)]	0.340	0.269	0.418
REVI	SQPAD-C [p(r0.66)]	0.317	0.258	0.380
REVI	QPAD	0.423	0.318	0.539
REVI	Naïve	0.270	0.233	0.300
SWTH	SQPAD-O [mean(p_ij)]	0.144	0.098	0.201
SWTH	SQPAD-O [p(r0.66)]	0.133	0.098	0.180
SWTH	SQPAD-C [mean(p_ij)]	0.150	0.105	0.203
SWTH	SQPAD-C [p(r0.66)]	0.143	0.108	0.194
SWTH	QPAD	0.206	0.131	0.300
SWTH	Naïve	0.123	0.099	0.152
TEWA	SQPAD-O [mean(p_ij)]	0.500	0.415	0.614
TEWA	SQPAD-O [p(r0.66)]	0.456	0.368	0.558
TEWA	SQPAD-C [mean(p_ij)]	0.554	0.450	0.680
TEWA	SQPAD-C [p(r0.66)]	0.492	0.400	0.613
TEWA	QPAD	0.813	0.647	1.021
TEWA	Naïve	0.299	0.265	0.340
WIWR	SQPAD-O [mean(p_ij)]	0.104	0.072	0.143
WIWR	SQPAD-O [p(r0.66)]	0.093	0.070	0.125
WIWR	SQPAD-C [mean(p_ij)]	0.111	0.075	0.157
WIWR	SQPAD-C [p(r0.66)]	0.100	0.073	0.135
WIWR	QPAD	0.149	0.092	0.224
WIWR	Naïve	0.082	0.063	0.100
WTSP	SQPAD-O [mean(p_ij)]	0.372	0.299	0.446
WTSP	SQPAD-O [p(r0.66)]	0.335	0.270	0.408
WTSP	SQPAD-C [mean(p_ij)]	0.448	0.378	0.534
WTSP	SQPAD-C [p(r0.66)]	0.406	0.348	0.478
WTSP	QPAD	0.610	0.494	0.747
WTSP	Naïve	0.332	0.292	0.382

8 Session info

- R version 4.4.2 (2024-10-31), aarch64-apple-darwin20
- Running under: macOS Sequoia 15.5
- Matrix products: default
- BLAS:
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
- LAPACK:
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib ;
LAPACK version 3.12.0
- Base packages: base, datasets, graphics, grDevices, methods, stats, stats4, utils
- Other packages: arrow 15.0.1, bSims 0.3-1, deldir 2.0-4, detect 0.5-0, dplyr 1.1.4, extraDistr 1.10.0, Formula 1.2-5, ggplot2 3.5.1, intrval 1.0-0, MASS 7.3-61, Matrix 1.7-1, mefa4 0.3-11, pbapply 1.7-4, qs 0.26.1, rconfig 0.3.0
- Loaded via a namespace (and not attached): assertthat 0.2.1, bit 4.0.5, bit64 4.0.5, cli 3.6.4, colorspace 2.1-1, compiler 4.4.2, digest 0.6.37, evaluate 1.0.3, farver 2.1.2, fastmap 1.2.0, generics 0.1.3, glue 1.8.0, grid 4.4.2, gtable 0.3.6, htmltools 0.5.8.1, isoband 0.2.7, knitr 1.50, labeling 0.4.3, lattice 0.22-6, lifecycle 1.0.4, magrittr 2.0.3, munsell 0.5.1, parallel 4.4.2, pillar 1.10.2, pkgconfig 2.0.3, purrr 1.0.2, R6 2.6.1, RApiSerialize 0.1.2, Rcpp 1.1.0, RcppParallel 5.1.7, rlang 1.1.5, rmarkdown 2.29, scales 1.3.0, stringfish 0.16.0, tibble 3.2.1, tidyselect 1.2.1, tinytex 0.56, tools 4.4.2, vctrs 0.6.5, viridisLite 0.4.2, withr 3.0.2, xfun 0.51, yaml 2.3.10